

Biochemistry Lec.5

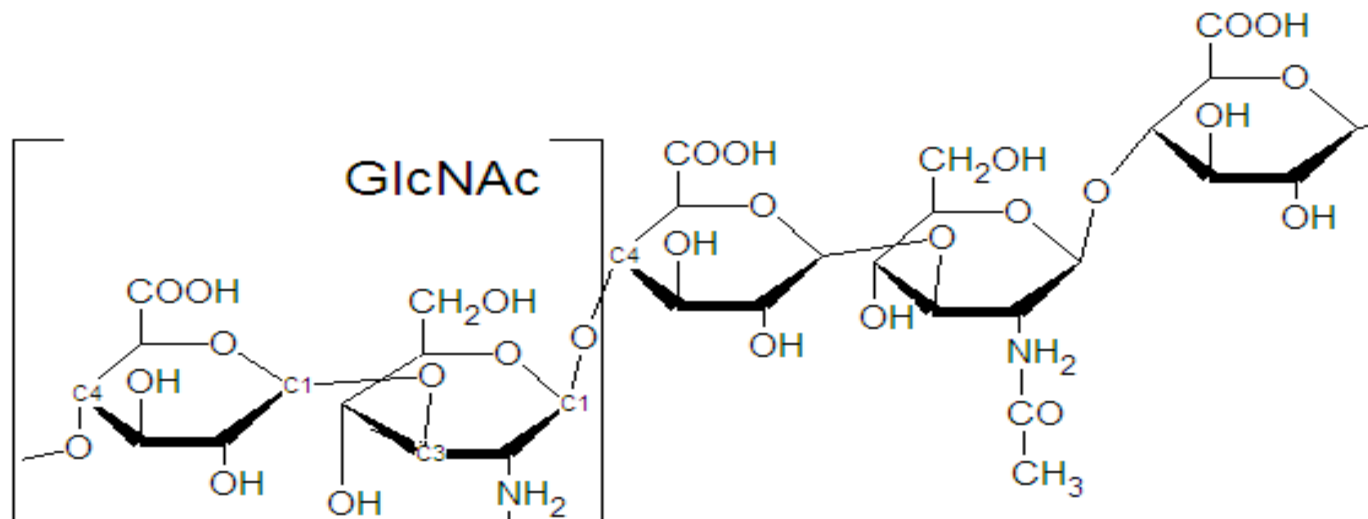
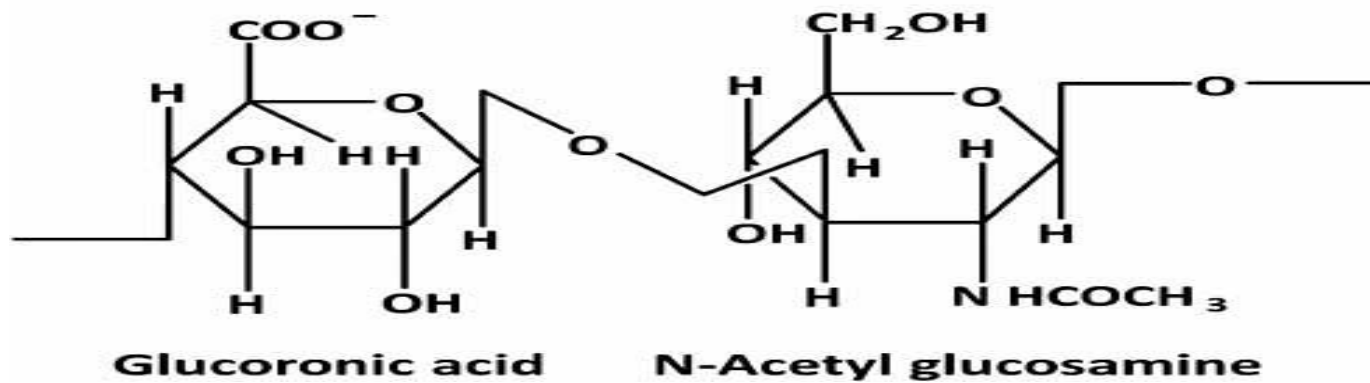
- Heteropolysaccharides
- GAGS



Hyaluronic acid

Its repeating unit consists of glucuronic acid and N-acetylglucosamine

Hyaluronic Acid (HA) Unit



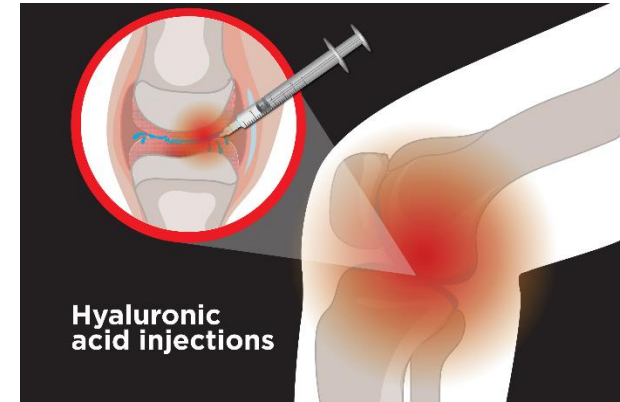
Hyaluronic acid

Biomedical importance:

- 1- Extracellular space of connective tissue as ground or cementing substances.
- 2- Synovial fluid of joint act as lubricant
- 3- vitreous humor of eye act as shock absorbent
- 3- Ovum for protection.

Enzyme “hyaluronidase “ hydrolyses hyaluronic acid.

- Hyaluronidase present in head of sperm , hydrolyzes the hyaluronic acid present on ovum which facilitates its penetration and fertilization.
- Snake venom is rich in hyaluronidase, thus snake bite hydrolyzes and liquifies the hyaluronic acid present in extracellular spaces of cells facilitate the spreading of toxin.
- Hyaluronidase is called spreading factor.



Heparin

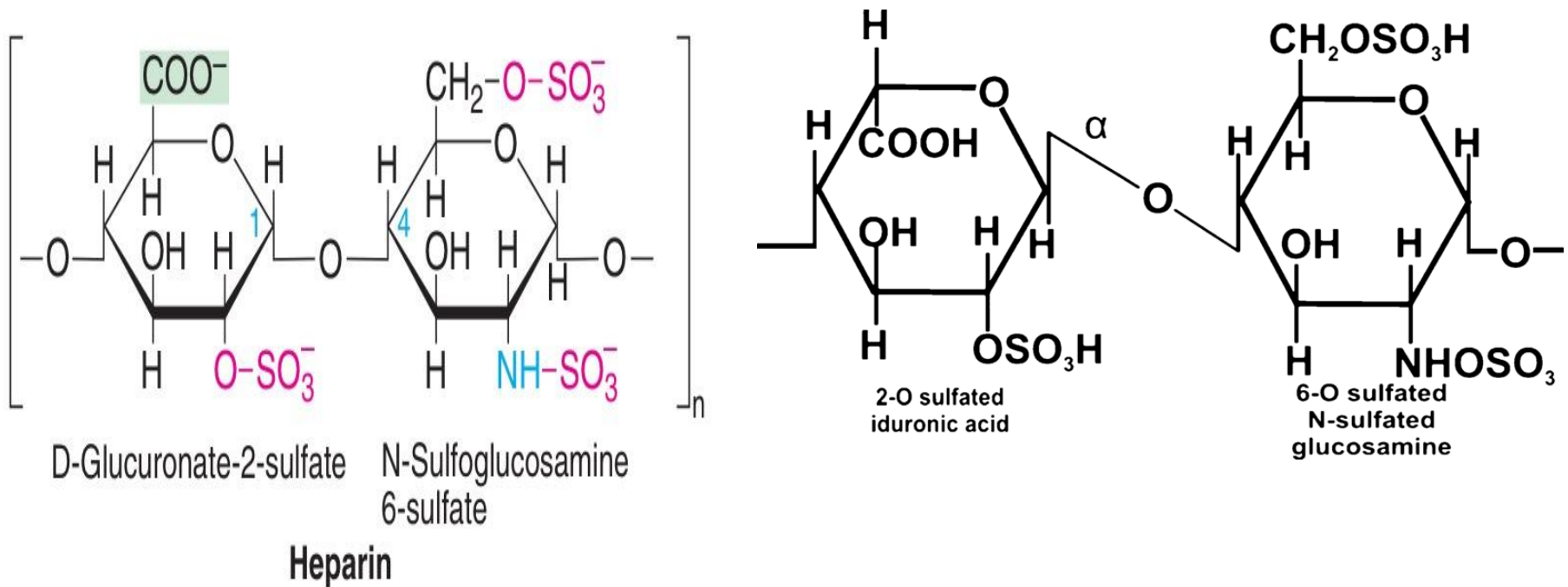
Its repeating unit consists of:

Glucuronic acid sulfate and glucosamine sulfate (minor)

Iduronic acid sulfate and glucosamine sulfate (major)

It is soluble highly sulfated GAG present in mast cell, blood vessel, lung, liver, spleen

It act anticoagulant



Heparan sulfate

Its repeating unit consists of:

Glucuronic acid sulfate and N-acetylglucosamine sulfate (major)

Iduronic acid sulfate and N-acetylglucosamine sulfate (minor)

- Heparan Sulfate present on extracellular cell surfaces or plasma membranes and serves as receptors
- Participate in cell growth, cell adhesion, and cell -cell communication.
- Heparan sulfate in basement membrane of kidney helps in determining charge selectiveness of Glomerular filtration.
- Heparan sulfate are also components of aorta, liver, fibroblasts, synaptic and vesicles.

Chondroitin sulfate-4 and -6

Its repeating unit consists of:

Glucuronic acid and N-acetylgalactosamine sulfate

- Chondroitin Sulfate is present in connective tissues-bones, cartilage, tendons .
- It gives mechanical strength, compressibility and support to connective tissues.
- Chondroitin sulfate present in cornea give over all shape to eye.

